Science and Sustainability Assignment 2: Massey News article

Student Name: John Lavack

Student ID number: 20001249

Course number and title: 247111 Science and Sustainability_2023_S1FS

Transform Trash to Treasure

Waste is a big problem. The very word is antithetical to sustainability. Particularly when the abundance of waste compromises the environment and future generations; counter to the definition of sustainability in the Brundtland report (World Commission on Environment and Development, 1987). According to environment.govt.nz (2021) 72% of NZ waste ends up in landfills. Massey's climate action plan (2019) specifies a goal to reduce waste to zero (currently 4% of Massey's emissions), partly by recycling/reusing waste and diverting it from the landfill. Can we transform part of the waste to something valuable, thereby avoiding the landfill? Enter the humble fungus.

New Zealand's fungi, which play a vital role in our ecosystems, are a mystery. It is estimated NZ has 22,000 endemic species, with 7,500 recorded (New Zealand Plant Conservation Network 2023). Fungi decompose things. Naturally, this has promise in waste management. Fungi already break down organic matter very well. They can also break down plastics. A 2013 study by Khalil et al has shown that white-rot fungi can decompose plastic polymers in garbage (polyethylene) given the right conditions (pH = 5.0, temperature = 30C), with only one promising strain. A later 2022 review of plastic degrading fungi by Ekanayaka et al indicates 395 such strains exist in 11 fungal phyla, and can degrade plastic under a variety of conditions cost effectively, with Eurotiomycete having the largest number. For which there are 28 observed species in NZ (iNatualist. 2023). These studies offer the potential of using fungi to transform waste into something of value in a way that aligns with Mātauranga Māori principles of whakapapa and mauri.

Whakapapa, translated as 'to place in layers' (Rameka, 2016) encourages one to find connection with their lineage and land. Informing relationships and seeing interdependence between all things, by layering these connections from the dawn of phenomenological time up to the individual. Sheldrake (2020) shows that Fungi connect nature in a layered way, via the wood-wide-web, as decomposers and recycling chemicals in the biogeochemical cycle. This cyclical process of matter and energy through an ecosystem surfaced by the idea of mauri. Where we have strong fungal networks we see mauri ora. Where these are damaged by litter and dumping we see mauri noho and where we see landfills we observe mauri rere. Using fungi to do what they are meant to do (decompose, recycle and rebalance) is an act of reestablishing mauri tau and mauri oho.

Using NZ fungi to target the waste issue, requires us to know what fungi there are. We know little about NZ fungi. Creating a fungi whakapapa can help us to find NZ's fungi decomposers. As with the two studies, at first little was known, but as more was learnt, more strains were uncovered. Partnering with kaitiaki such as the Department of Conservation, Iwi and Hapū to collect and cultivate the fungi. Then developing a process to decompose the plastic and other, non-electronic waste into compost. An extreme example of this transformation is the fungi Mutarium turning plastic waste to food (Howarth, 2014). Much less extreme would be the reintegration of the compost as mauri (life force) to the land. Thereby reducing the emissions of Massey as a result, and creating a truly sustainable waste cycle.

References

Ekanayaka, A. H., Tibpromma, S., Dai, D., Xu, R., Suwannarach, N., Stephenson, S. L., Dao, C., Karunarathna, S. C. (2022). *A Review of the Fungi That Degrade Plastic. Journal of Fungi (Basel)*, *8(8)*, 772. doi: 10.3390/jof8080772.

Howarth, D. (2014). *Fungi Mutarium turns waste plastic into edible treats.* https://www.dezeen.com/2014/12/10/livin-studio-katharina-unger-fungi-mutarium-recycle-plastic-food/

iNaturalist. (2023, April 25). *Eurotiomcetes Observations in New Zealand.* https://www.inaturalist.org/observations?place_id=6803&taxon_id=117868

Kahlil, M. I., Ramadan, N. A., Albarhawi, R. K. (2013). *Biodegradation of polymers by Fungi isolated from Plastic garbage and assay for Optimum Condition for growth.* https://www.researchgate.net/publication/260780595_Biodegradation_of_polymers_by_Fungi_isolated_from_Plastic_garbage_and_assay_for_Optimum_Condition_for_growth

Massey University. (2019, July). *Climate Action Plan 2020-2030. https://www.massey.ac.nz/massey/fms/sustainability/documents/Strategy/Draft%20Climate%20Action%20Plan%202020-2030.pdf*

Ministry of the Environment. (2021, September 9). *Estimates of waste generated in Aoteroa New Zealand.* https://environment.govt.nz/facts-and-science/waste/estimates-of-waste-generated/

New Zealand Plant Conservation Network. (2023). *Fungi.* https://www.nzpcn.org.nz/flora/fungi/

Rameka, L. (2016). *Whakapapa: A Māori Way of Knowing and Being in the World. Encyclopedia of Educational Philosophy and Theory.* https://doi.org/10.1007/978-981-287-532-7_172-1

Sheldrake, M. (2020). *Entangled Life: How Fungi Make Our Worlds, Change Our Minds and Shape Our Futures*. Vintage Publishing.

World Commission on Environment and Development. (1987). *Our Common Future.* Oxford University Press

…

Assignment 2 – self assessment check list. Mark your self-assessment for each achievement criteria with an x in the highlighted table cell.

Achievement criteria	**D**	**C**	**B**	**A**
2.1.1 – Use credible information sources.	No credible sources are used	Credible sources used but lacking scholarly peer-reviewed information	At least one scholarly peer-reviewed source is used	At least one scholarly peer-reviewed source used that has been discussed in the context of a broader range of credible information
				x

2.1.2 – Summarise a source of information in your own words, identifying key points relevant to the topic.	Misses many key points, OR no restructuring of knowledge, OR inclusion of irrelevant information, OR inaccurate	Captures most of the main points with attempt at restructuring, OR captures all the main points but without substantively restructuring	Key concepts are captured with evidence of restructured knowledge – e.g. own words, own structure	Key concepts are communicated clearly and succinctly AND ideas are fully restructured from source materials

	restructuring and consistently poor paraphrasing			
				x
2.2.1 – Use evidence appropriately to substantiate ideas.	Statements consistently unsupported, OR incorrect attributions made	Few unsupported statements BUT some errors in citation and/or referencing, or omissions from reference list	Consistent attribution of sources including quotes, AND complete reference list	Consistent and accurate attribution of sources including quotes, AND complete and properly formatted reference list
				x

2.2.2 – Use appropriate style and structural conventions for informal writing.	Lacks appropriate sections and shows no logical progression in structure	Contains appropriate sections, but some inconsistencies in structure OR Logical progression of ideas, but lacking formal structure	Logically structured and appropriate sections with some attempts at creating connections between sections	Well-structured and well-connected sections with clear logical progression of structure
				x

2.2.3 – **Convey relevant ideas to the general public.**	Language, tone and information selected shows lack of awareness of target audience	Awareness of audience inconsistently visible through language and tone	Appropriate use of language and tone showing consistent awareness of audience	Consistent awareness of audience with information presented in manner designed to effectively engage general public
				x

2.3.1 – Relate Mātauranga Māori concepts with western science in relation to a sustainability topic.	Inaccurate explanations of western science or Mātauranga Māori, OR inaccurate application to sustainability topic, OR no/insufficient attempt to relate western science and Mātauranga Māori concepts	Accurate explanation of Mātauranga Māori concepts, BUT limited linkage to science and/or application to sustainability topic	Accurate explanation of western science and Mātauranga Māori perspectives in relation to sustainability topic	Accurate explanation of western science and Mātauranga Māori perspectives in relation to sustainability topic that explores the opportunity for partnership between science and Te Ao Māori
				x

2.5.1 – Reflect on the wider concept of sustainability and how this is relevant to the context of your topic.	Significant misconceptions or inaccuracies in explanation or application to topic	Mostly accurate explanation but connections/ application to topic not well developed	Accurate explanation appropriately related to topic	High level explanation of sustainability and situation shows connection between own understanding, concepts of sustainability discussed in course, and scientific literature
				x